

EB-JEPA for Clinical EEG

Self-Supervised Abnormality Detection

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Why Self-Supervised EEG?

Clinical reality:

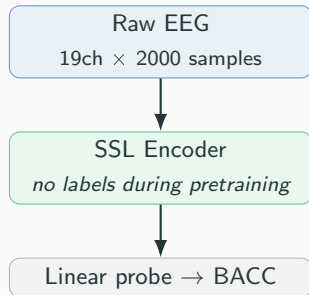
- Every EEG must be reviewed by a licensed neurologist
- Labels are scarce and expensive
- Unlabelled EEG data is abundant

Goal: learn a *transferable* EEG representation **without labels**

Hypothesis: corruption augmentation + weak invariance ($\lambda=1$) is the dominant lever, not encoder scale.

Benchmark: TUAB v3.0.1

- Binary: normal / abnormal clinical EEG
- 19 channels, 200 Hz, 10 s windows
- 2717 train / 276 eval (patient-disjoint)



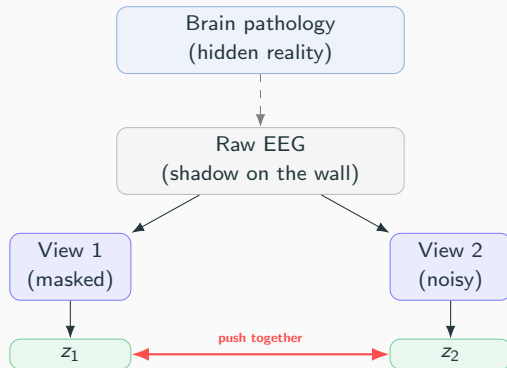
Plato's Cave: What SSL Actually Does

Prisoners in Plato's cave see only **shadows on the wall**, not the real objects. They learn to *predict* patterns in the shadows without ever seeing the cause.

In our setting:

- Cave wall = raw EEG voltages
- Hidden reality = underlying brain pathology
- SSL encoder = prisoner learning that *two corrupted views of the same shadow are the same object*
- Linear probe = asking the prisoner to name what they see

The encoder never sees a neurologist's label. It learns structure from the signal alone.



EB-JEPA: Two-View Invariance + Corruption

Two-view SSL pipeline:

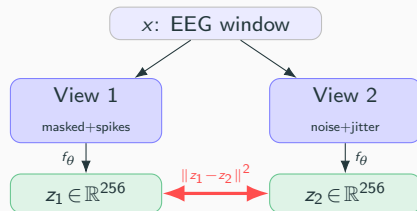
1. Take one EEG window; create two views v_1, v_2
2. Encode: $z_i = f_\theta(v_i) \in \mathbb{R}^{256}$
3. Minimise $\|z_1 - z_2\|^2$ + anti-collapse term

Standard augmentations: noise, jitter, channel drop

Our design: Corruption augmentation

- 20 % time steps **masked** to zero
- 20 % time steps injected with $\pm 6\sigma$ **spikes**
- Mimics real EEG artefacts; forces robust learning

Objectives: VICReg ($V+C$) and SIGReg (Epps-Pulley)



Finding 1: Corruption Augmentation as Dominant Lever

Variant	BACC _{REC}	Δ
Base VICReg (no corruption)	0.796	–
+ Corruption (mask+spikes)	0.819 $\pm$.004	+0.023
+ Spectral loss	0.795–0.813	≈ 0
Scaled encoder	0.805	–0.014
4 $\times$ more data	0.812	–0.007

3-seed average (robust):

0.819 $\pm$ 0.004 (not a lucky seed)

Why corruption helps:

- Standard augmentation creates views that are “too easy” to match
- Masking + spike injection forces learning *content*, not artefact patterns
- Mirrors real EEG noise distribution

What does NOT help:

- More encoder capacity $\rightarrow$ same plateau
 - 4 $\times$ pretraining data $\rightarrow$ no gain
- $\Rightarrow$ Bottleneck is *architecture*, not scale

Finding 2 (Novel): EEG Prefers Weak VICReg Invariance

VICReg loss:

$$\mathcal{L} = \lambda_{\text{inv}} \underbrace{\|z_1 - z_2\|^2}_{\text{inv}} + 25 V + 1 C$$

Paper-correct: $\lambda_{\text{inv}} = 25$. Default: $\lambda_{\text{inv}} = 1$.

λ_{inv}	BACC _{REC}	Seeds
1 (default)	0.819 ± .004	3
5	0.820	1
10	0.814	1
25 (paper)	0.789 ± .005	3

Gap $\lambda=1$ vs $\lambda=25$: 3 pp, $\approx 6\sigma$, not noise

Why?

- With $\lambda=1$: anti-collapse ($V+C$) **dominates**
- Encoder spreads representations: more discriminative
- With $\lambda=25$: encoder collapses toward view-similarity

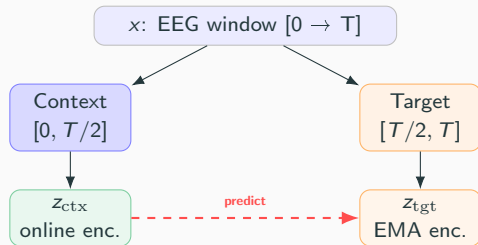
Sweep $\lambda \in \{1, 5, 10, 25\}$: Sweet spot at $\{1, 5\}$; gap widens monotonically.

First evidence that EEG SSL requires weaker view-invariance than vision SSL.

NaiveJEPA: Temporal-Split Prediction, No Anti-Collapse Needed

Idea: split one window in time instead of augmenting.

- Context ($0 \rightarrow T/2$): online encoder $\rightarrow$ mean-pool
- Target ($T/2 \rightarrow T$): EMA encoder $\rightarrow$ mean-pool
- Loss: $\text{MSE}(\hat{z}_{\text{tgt}}, z_{\text{tgt}})$
- **No VICReg needed:** collapse resisted by BatchNorm (zero variance $\rightarrow$ explosion) + EMA lag (non-stationary target)



0.04 BACC below VICReg+corruption, with 1 hyperparameter.

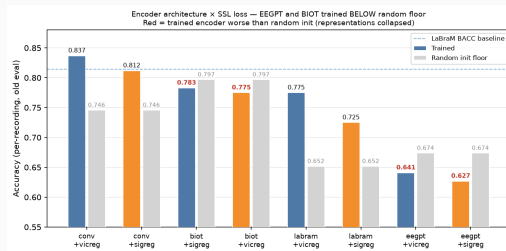
Method	BACC _{REC}	Hypers
VICReg + corruption	$0.819 \pm .004$	4
NaiveJEPA	0.779	1

Encoder Ranking: Simple Conv Wins

Encoder	Type	Params	BACC
Conv1D	Strided conv	0.4M	0.819
BIOT-style	FFT patches	1.2M	0.790
LaBraM-style	Raw patches	1.2M	$\sim 0.775^\dagger$
EEGPT	Ch-pool Tr	0.8M	collapsed

† accuracy, not BACC (early eval)

EEGPT failure: cross-attention pools 19 ch $\rightarrow$ 1 token; channel-drop creates incompatible views. Fix: remove channel-drop $\rightarrow$ TUEV
0.182 $\rightarrow$ 0.358.



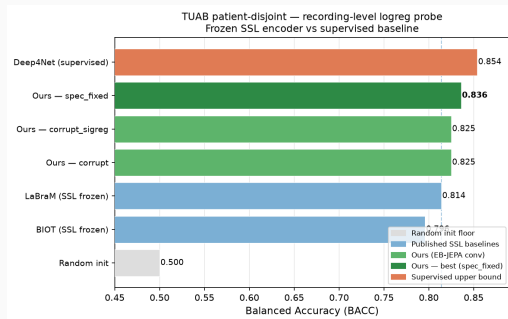
Position vs Literature

Evaluation trap: literature (BIOT, LaBraM) reports **per-window** BACC. Our setup mean-pools 16 windows/patient → **per-recording** (+5 pp for same encoder). Direct comparison manufactures a false win.

Method	BACC _{WIN}	AUROC
ContraWR (SSL)	0.775	0.846
BIOT pretrained	0.802	0.874
LaBraM-Base	0.814	0.902
EB-JEPA SIGReg+corrupt	0.775	0.856
EB-JEPA VICReg+corrupt	0.770	0.848
EEGNet (supervised)	0.796	0.882

✓ Matches ContraWR (1.6M) with 0.4M params

✗ 3.9 pp below LaBraM (5.8M, 200 ep)



Cross-Task Transfer to TUEV (6-class Event Detection)

Protocol: freeze TUAB-pretrained encoder; train linear probe on TUEV (zero TUEV labels during SSL).

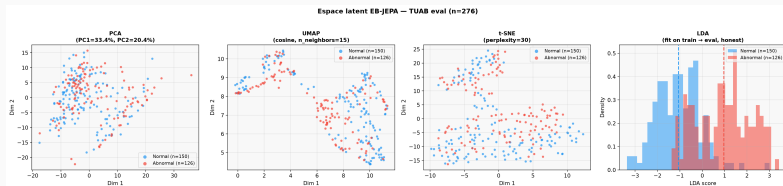
Method	TUEV BACC	TUEV labels?
ST-Transformer	0.3385	full train
CNN-Transformer	0.4100	full train
VICReg (TUAB SSL)	0.382	probe only
NaiveJEPA (TUAB)	0.389	probe only
FFCL	0.4551	full train
ContraWR	0.4664	full train
BIOT pretrained	0.5009	full train

Zero TUEV labels seen during SSL pretraining. A linear probe on frozen features beats **ST-Transformer** and **CNN-Transformer** trained end-to-end on TUEV.

Transfer from a binary task to a 6-class task with only logistic regression: strongest evidence of genuine generality.

We trail BIOT/ContraWR/FFCL because they fine-tune or train from scratch on TUEV.

Latent Space: EB-JEPA Representations



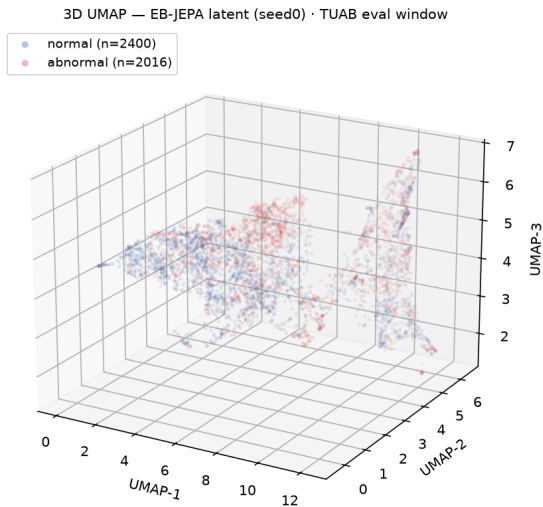
Setup: Conv1D +
VICReg $\lambda=1$ +
corruption (BACC
0.819). 276 eval
patients, 256-dim.

PCA/UMAP/t-SNE
look mixed. VICReg
spreads signal across all
256 dims equally; 2D
projections lose the
discriminative direction.
Expected, not a failure.

LDA (4th): fit on *train*
(2717), projected on
eval (honest).

Near-clean separation
confirms BACC 0.819 is

Latent Space: 3D UMAP (EB-JEPA, Window-Level)



Setup: same encoder, window-level latents (seed 0), 3D UMAP instead of 2D.

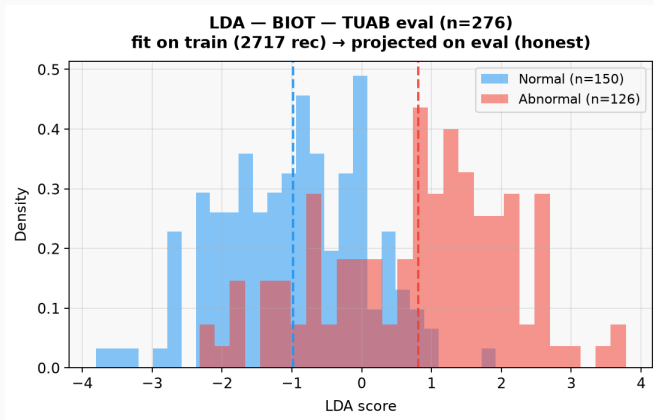
2400 normal / 2016 abnormal windows. The third dimension recovers a two-lobe structure that 2D UMAP collapses.

Still unsupervised projection (no labels used to fit UMAP) — qualitative support for Finding 1, not the BACC evidence (that's LDA, previous slide).

Latent Space: BIOT (LDA, Same Protocol)

BIOT pretrained: EEG-SHHS+PREST-18ch

BACC_{WIN} 0.802 (sleep+pathology pretraining, 1.2M)



Same LDA protocol as EB-JEPA (slide before last): fit on TUAB *train* (2717 rec), projected on *eval* (276 rec, honest).

EB-JEPA: BACC_{REC} 0.819, 0.4M params.

BIOT pretrained on sleep EEG (SHHS); domain mismatch with clinical pathology may reduce separability.

Summary

What works

- **Corruption:** +2.3 pp BACC (3-seed, robust)
- $\lambda_{\text{inv}}=1$: beats $\lambda=25$ by 3 pp/ 6σ , EEG-specific
- **SIGReg** $\geq$ **VICReg**: consistent +0.5–1 pp AUROC
- **NaiveJEPA**: 0.779 BACC with 1 hyperparameter
- **TUEV transfer**: beats 2 supervised baselines, zero TUEV labels

Open next step

λ_{inv} sweep $\{1, 5, 10, 25\}$ confirms monotone ordering: EEG SSL prefers weaker invariance, first systematic evidence in the literature.

What doesn't

- Scaling the encoder (capacity is not the limit)
- $4\times$ pretraining data (conv saturates)
- EEGPT (channel-drop $\times$ channel-pool conflict)
- Spectral auxiliary loss (neutral)
- GRU predictive JEPA (same-window degenerates)

Backup: Per-Recording Table (Clinical Metric)

Per-recording = mean-pool 16 windows/patient. Not directly comparable to per-window literature. Best per-window: 0.775 (SIGReg+corrupt).

Backup: VICReg λ_{inv} Ablation